

The register() View Function

The register() view function needs to display a blank registration form when the registration page is first requested, and then process completed registration forms when they're submitted. When a registration is successful, the function also needs to log the new user in. Add the following code to *accounts/views.py*:

```
accounts/
views.py  from django.shortcuts import render, redirect
          from django.contrib.auth import login
          from django.contrib.auth.forms import UserCreationForm

          def register(request):
              """Register a new user."""
              if request.method != 'POST':
                  # Display blank registration form.
                  ❶ form = UserCreationForm()
              else:
                  # Process completed form.
                  ❷ form = UserCreationForm(data=request.POST)

                  ❸ if form.is_valid():
                      ❹ new_user = form.save()
                      # Log the user in and then redirect to home page.
                      login(request, new_user)
                      ❺ ❻ return redirect('learning_logs:index')

              # Display a blank or invalid form.
              context = {'form': form}
              return render(request, 'registration/register.html', context)
```

We import the render() and redirect() functions, and then we import the login() function to log the user in if their registration information is correct. We also import the default UserCreationForm. In the register() function, we check whether we're responding to a POST request. If we're not, we make an instance of UserCreationForm with no initial data ❶.

If we're responding to a POST request, we make an instance of UserCreationForm based on the submitted data ❷. We check that the data is valid ❸—in this case, that the username has the appropriate characters, the passwords match, and the user isn't trying to do anything malicious in their submission.

If the submitted data is valid, we call the form's save() method to save the username and the hash of the password to the database ❹. The save() method returns the newly created user object, which we assign to new_user. When the user's information is saved, we log them in by calling the login() function with the request and new_user objects ❺, which creates a valid session for the new user. Finally, we redirect the user to the home page ❻,